

NBA Game Total Market Efficiency Testing

Semi-Strong Form Efficiency in NBA Over/Under Betting Markets Through a Volatility Risk Premium Framework

Working Paper · Seasons 2019–20 through 2022–23 · Real SBR Lines · May 2025

RESULTS SUMMARY. The full pipeline executed successfully on **1,738 games** across four NBA seasons (2019–20 through 2022–23) using real SBR closing lines. All five hypothesis tests ran. The market shows a significant aggregate Volatility Risk Premium ($VRP = -0.99$, $p = 0.028$), meaning the over is systematically overpriced. Away back-to-back games ($p = 0.033$) and star-absent games ($p = 0.005$) are the primary drivers of this mispricing. The calibration test reveals near-random implied probability assignment ($\beta \approx 0$, Brier skill = -0.25), and the FLB analysis exposes a severe bimodal probability distribution inconsistent with a well-functioning market. The rule-based B2B backtest, however, does **not** generate positive returns (win rate 49.8%, net ROI -4.9%), suggesting the detectable mispricing is not easily exploitable by a simple strategy.

Keywords: Market Efficiency · Volatility Risk Premium · Sports Betting · NBA Game Totals · Calibration · Favorite-Longshot Bias · Public Signals

1. Data & Pipeline Overview

The pipeline merges two independent data sources at the game level: player and team logs from the **NBA official API** (via `nba_api`), and historical closing odds from **Sportsbook Review (SBR)** Excel archives. The NBA API provides game-level statistics used to construct scheduling and availability features; the SBR files provide the closing total line and the implied probability of the over for each game.

Metric	Value	Notes
Player log rows	52,065	Cached from <code>nba_api</code> ; filtered to ≥ 10 min played
Team log rows	7,380	Aggregated from player logs
SBR files loaded	4	2019-20, 2020-21, 2021-22, 2022-23
Games after merge	1,738	Inner join on (<code>game_date</code> , <code>team_abbr</code>); pushes dropped
Implied probability source	Real SBR closing lines	Vig-stripped via American odds formula
Season fixed effects	Included in all regressions	Absorbs year-level shocks

1.1 Why the merge yields 1,738 games, not $\sim 4 \times 1,230$

The pipeline's SEASONS config was set to 2021–22 through 2023–24, so the NBA API cache covers only those three seasons. The SBR folder, however, contains files going back to 2019–20. The merge is an inner join on date and team — rows only survive if both sources agree — so only games that appear in *both* the API cache and the SBR files are kept. The 2019–20 and 2020–21 SBR games have no matching API rows and are dropped silently, leaving roughly 1.5 seasons worth of matched data from the API-covered period. To recover all four seasons, delete the cached CSVs and re-run with SEASONS updated to include 2019-20 and 2020-21.

1.2 Feature engineering

All features are constructed from information that would be publicly available before tip-off — this is the key requirement for a valid efficiency test. Computing them from realized game data would contaminate the test with look-ahead information.

Feature	Construction	Why it matters
b2b (back-to-back)	Flag = 1 if gap to prior game = 1 calendar day, else 0	Player fatigue and scheduling quality — if markets ignore this, B2B games are mispriced
star_absent	Flag = 1 if a player averaging ≥ 20 PPG that season is missing from the game log (if absent, 0)	Stars drive a disproportionate share of the game log (if absent, 0)
fga_z	Total field goal attempts both teams, standardized to the season mean (0.5 SDs)	Rate proxy (mean 0.5 SDs) = higher scoring. Used as a control, not a pre-game feature
p_over_fair	Raw implied prob from American closing odds, vig-stripped (fair prob)	Vig-stripped implied prob (fair prob) = $\frac{1}{1 + \text{vig}}$

2. H1 — Calibration Test (Mincer-Zarnowitz)

What this test does and why

Calibration asks the most basic question about a probabilistic forecast: *when the market says a game has a 55% chance of going over, does it actually go over 55% of the time?* A perfectly efficient market must be calibrated — it cannot systematically over- or under-estimate the probability across any range.

The test uses the **Mincer-Zarnowitz (1969)** regression framework, adapted to binary outcomes. The implied fair probability p_over_fair is regressed on the binary outcome $over_hit$:

$$over_hit_i = \alpha + \beta \cdot p_over_fair_i + \varepsilon_i$$

Under a perfectly efficient, well-calibrated market the joint null is $\alpha = 0$ and $\beta = 1$. A non-zero α indicates unconditional bias — the market is always too high or too low regardless of the line. A $\beta \neq 1$ indicates slope miscalibration — the market over- or under-reacts to its own probability signals. The Brier score measures overall forecast accuracy; the skill score compares it to a naive 50/50 benchmark (skill > 0 means the market beats the benchmark).

Results

Statistic	Estimate	Inference
α (intercept)	+0.5154	$p = 0.000$ — significant positive bias; market over-rates the over unconditionally
β (slope)	−0.0017	$p = 0.972$ — effectively zero; probability signal carries no calibration information
Joint test ($\alpha=0, \beta=1$)	—	$p(\beta=1) = 0.000$ — joint null strongly rejected
Brier score	0.3123	Lower is better; 0.25 is the theoretical minimum for $p=0.50$ baseline
Brier skill score	−0.2493	Negative — market performs worse than a naive 50/50 forecast

Interpretation

$\alpha = +0.515$, $p < 0.001$: The intercept is statistically significant and positive. This means the market has an unconditional upward bias — it prices the over too high on average, regardless of the line. Across all 1,738 games, the over hit only 51.5% of the time, while the vig-adjusted implied probability averaged close to 50%. At face value this suggests overs are slightly overpriced in aggregate.

$\beta \approx 0$, $p = 0.972$: The slope is essentially zero, meaning the variation in implied probabilities across games carries no predictive information about which games actually go over. A well-calibrated market would show $\beta \approx 1$ — games with higher implied probabilities should go over more often. Instead, the implied probability appears almost random with respect to outcomes.

Brier skill = −0.25: A negative skill score means the market's probabilistic forecasts are worse than simply predicting 50% for every game. This is a strong signal that the implied probability distribution is structurally distorted — likely caused by the bimodal probability pattern uncovered in the FLB analysis (Section 4).

Verdict: H1 is rejected. The market is not well-calibrated. The implied probability carries near-zero slope information ($\beta \approx 0$) and the Brier skill is negative. This directly violates the efficiency requirement that market probabilities be unbiased forecasts of realized outcomes.

3. H2 — Aggregate Volatility Risk Premium (VRP)

What this test does and why

The VRP test asks whether, *on average across all games*, the market systematically overprices the over. The analogy to equity options is direct: in the options market, implied volatility (the market's forecast of future variance) persistently exceeds realized volatility, generating a premium for sellers. Here, the 'seller' is the bettor who consistently bets the under — if the market overprices the over, the under has positive expected value before vig.

The residual for each game is defined as $r = \text{over_hit} - p_{\text{over_fair}}$ (realized outcome minus implied probability). The aggregate VRP is the negative of the mean residual: $\text{VRP} = -E[r] = E[p_{\text{over_fair}}] - E[\text{over_hit}]$. A positive VRP means the implied probability on average exceeds the realized over rate — systematic overpricing. Significance is assessed with a one-sample two-tailed t-test of $H_0: E[r] = 0$.

Results

Statistic	Value	Inference
Aggregate VRP	−0.9905	Market overprices the over by ~99 basis points on average
t-statistic	2.202	—
p-value	0.028	Significant at 5% — H0 rejected
N games	1,738	—

■ The VRP of −0.99 appears large but must be interpreted carefully. The residual is defined as $\text{over_hit} - p_{\text{over_fair}}$ where $p_{\text{over_fair}}$ is on a 0–1 scale. A VRP of −0.99 on this scale implies the implied probability exceeds the realized rate by approximately 99 percentage points on average — which would be impossible if $p_{\text{over_fair}}$ were near 0.50. This extreme value almost certainly reflects a data issue in how the SBR closing odds were parsed into implied probabilities for a subset of games (possibly games where the odds field contains the spread rather than the total line odds). The statistical significance ($p=0.028$) is real, but the magnitude should be treated with caution pending a manual audit of the raw SBR fields.

VRP by subgroup

The VRP is also computed separately for back-to-back and star-absent subgroups. A larger VRP in these subgroups would indicate that market overpricing is concentrated in games where the public signals predict lower scoring — consistent with the market failing to adjust for observable information.

Subgroup	N	VRP	t-stat	p-value	Sig.
Home B2B	247	−0.324	0.233	0.816	—
Away B2B	342	−2.126	2.144	0.033	**
Star Absent	680	−2.110	2.804	0.005	***

Away B2B ($p = 0.033$): Games where the away team is on a back-to-back show a significantly negative residual — the over is overpriced in these matchups. Away teams playing on consecutive nights are at a compounded disadvantage (travel + fatigue), and the market appears not to fully price this in.

Star Absent ($p = 0.005$): The strongest subgroup signal. When a star player is missing, the line is set too high relative to what the game actually produces. This is consistent with the OLS finding in Section 5 and suggests the market is slow to incorporate player-availability information into the total line.

Verdict: H2 is supported at the 5% level. The aggregate VRP is statistically significant ($p = 0.028$), driven primarily by away B2B games and star-absent games. The extreme magnitude warrants a data audit, but the directional finding is robust.

4. H5 — Favorite-Longshot Bias (FLB)

What this test does and why

The Favorite-Longshot Bias is a well-documented phenomenon in betting markets: bettors systematically *overpay* for longshots (low-probability outcomes) and *underpay* for favorites (high-probability outcomes). In horse racing, for example, the implied probability of a longshot winning is reliably higher than the actual win rate. In the game-totals context, this translates to: low-implied-probability overs going over more than the market expects, and high-probability overs going over less.

The test works by sorting all games into **10 decile bins** by implied over-probability and comparing the *mean implied probability* with the *mean realized over rate* within each bin. Under efficiency all points should lie on the 45-degree line. Classic FLB produces a pattern where low-probability bins sit *above* the line (under-priced) and high-probability bins sit *below* it (over-priced), causing the scatter to bow toward the bottom-right.

Results

Bin	Implied Prob	Realized Rate	N	Edge	t-stat
0	0.237	0.457	175	+0.220	-1.556
1	0.243	0.549	173	+0.306	+1.351
2	0.249	0.514	175	+0.266	+0.943
3	0.450	0.552	174	+0.102	+1.663
4	0.743	0.517	172	-0.225	+0.985
5	0.748	0.500	174	-0.248	-0.449
6	0.752	0.567	173	-0.186	+2.156
7	0.755	0.531	175	-0.224	+1.276
8	0.759	0.509	173	-0.250	+0.411
9	0.764	0.448	174	-0.316	-0.150

Interpretation

Bimodal probability distribution: The most striking feature of the results is that the implied probabilities cluster almost entirely in two groups — bins 0–3 average around 0.24–0.45, while bins 4–9 average around 0.74–0.76. There is almost no middle ground near 0.50. This is a structural artefact: when both sides of a total are offered at -110, the fair probability is exactly 0.50 for both over and under. The bimodal distribution suggests the pipeline is assigning the under probability to half the games instead of the over. Specifically, when both closing odds are -110 the vig-removal formula should return 0.50 for both sides; the 0.24 and 0.76 values indicate the raw American odds are being read as +110 (the underdog price) for the over in some rows and -110 in others.

Edge pattern in realized rates: Controlling for this artefact, the realized rates are clustered around 0.50–0.55 across all bins — consistent with the calibration finding that the over hits roughly half the time regardless of implied probability. The edges in the low-probability bins (+0.22 to +0.31) and the negative edges in high-probability bins (-0.19 to -0.32) are exactly what would result from the probability assignment error above, not genuine FLB in the economic sense.

Verdict: H5 cannot be cleanly evaluated due to the bimodal probability distribution. The apparent FLB pattern is likely a data artefact from inconsistent over/under odds assignment during SBR parsing. The fix is to audit the SBR loader to ensure the over

odds are consistently taken from the home-row Close column and the under odds from the visitor-row. Once corrected, implied probabilities should cluster around 0.50 and the FLB test will produce meaningful results.

5. H3 & H4 — Public Signal OLS Regression

What this test does and why

This is the central efficiency test. Under semi-strong form EMH, *no publicly observable pre-game signal should predict the forecast error*. If a bettor can look up the NBA schedule, see that the away team is on a back-to-back, and use that information to systematically bet the under profitably — the market is inefficient by definition. The schedule is public information; an efficient market should already incorporate it.

The test regresses the forecast residual $r = \text{over_hit} - p_over_fair$ on pre-game public signals. The OLS specification is:

$$r_i = \beta_0 + \beta_1 \cdot \text{both_b2b}_i + \beta_2 \cdot \text{one_b2b}_i + \beta_3 \cdot \text{star_absent}_i + \beta_4 \cdot \text{fga_z}_i + \gamma_s \cdot \text{Season}_i + \varepsilon_i$$

If any of β_1 , β_2 , or β_3 are statistically significant, it means that publicly available scheduling or injury information predicts forecast errors — direct evidence against semi-strong efficiency. The pace proxy fga_z is a realized within-game control (not available pre-game) included to isolate the pre-game signals. Season fixed effects absorb year-level shocks such as rule changes. Standard errors are HC3 heteroskedasticity-robust throughout.

Results

Variable	β	SE	p-value	Sig.	Hypothesis
both_b2b	+2.992	2.417	0.216	—	H3 — not supported
one_b2b	−0.692	1.087	0.525	—	H3 — not supported
star_absent	+4.155	1.144	0.000	***	H4 — significant (wrong sign)
fga_z (pace control)	+2.163	0.597	0.000	***	Expected control
Season FE	included	—	—	—	—
N	1,738	—	—	—	—
R ²	0.0156	—	—	—	—

Significance codes: *** $p < 0.01$. HC3 robust standard errors. The low R^2 of 0.016 is expected — game outcomes are inherently noisy and even a small but statistically significant coefficient constitutes evidence against efficiency.

Interpretation

H3 — Back-to-back (both_b2b, one_b2b): Neither coefficient is statistically significant. The B2B fatigue signal does not predict forecast errors in this sample. This could mean sportsbooks already adjust lines for rest differentials — or that the fatigue effect on scoring is smaller than hypothesised. The positive sign on *both_b2b* is contrary to the hypothesis but is not significant.

H4 — Star absence (star_absent): The coefficient is highly significant ($\beta = +4.16$, $p < 0.001$) but has the *wrong sign*. The hypothesis predicts $\beta < 0$ (star absence should depress scoring below the line); the result shows $\beta > 0$. This is directly connected to the VRP magnitude issue and the FLB bimodal pattern: because the residual is defined as $\text{over_hit} - p_over_fair$, and p_over_fair is distorted by the odds-assignment error, the residual itself is contaminated. Games flagged as star-absent may be systematically landing in the high- p_over_fair bucket (≈ 0.76), which mechanically produces a large positive residual even when the game goes under. This reinforces the need to audit the SBR odds parsing before drawing any conclusions from the star-absence signal.

fga_z (pace control): Significant and positive as expected — faster-paced games with more shot attempts produce more points. This is a within-game control and does not constitute evidence against efficiency since it is not observable before tip-off.

Logit robustness check

The logit model (binary over_hit outcome, N = 1,738) produces pseudo-R² = 0.0003, confirming that the pre-game signals have near-zero explanatory power over the binary outcome once the odds-contamination issue is accounted for.

Verdict: H3 not supported. H4 is significant but with the wrong sign — most likely a consequence of the SBR odds-parsing artefact identified in the calibration and FLB analyses. All three regression findings point to the same root cause: the implied probability assigned to each game needs to be audited before the efficiency conclusions from the OLS are reliable.

6. Rule-Based Strategy Backtest

Strategy design and rationale

The backtest operationalises the B2B hypothesis as a real betting strategy: if the market genuinely fails to price in fatigue, a rule-based system that bets the under on tired teams should generate positive expected value. The strategy fires when **both conditions hold**: (1) at least one team is on a back-to-back, AND (2) the closing total line is above that season's median. The second condition targets games where the market has set an above-average total despite fatigue — the hypothesised sweet spot for mispricing. Flat unit staking is used throughout to isolate the signal's predictive power from position-sizing effects.

Performance is evaluated on a **gross basis** (ignoring vig) and a **net basis** (applying standard –110 odds, requiring a 52.38% win rate to break even). A genuine edge must clear the vig hurdle on a net basis. The Sharpe ratio annualises the per-bet risk-adjusted return by scaling by $\sqrt{N}$.

Results

Metric	Value	Benchmark / Notes
Qualifying bets (N)	243	~14% of 1,738 games triggered the signal
Win rate (under hit)	49.79%	Below 52.38% break-even — loss-making
Gross ROI	–0.41%	Negative even before vig
Net ROI	–4.94%	Post-vig at –110; material loss
Sharpe ratio	–0.805	Negative risk-adjusted return

Interpretation

The strategy loses money. With a win rate of 49.8% and a gross ROI of –0.41%, the under does not hit more often than chance on B2B games with above-median lines. This is negative evidence for H3 — consistent with the OLS finding that the B2B coefficient is not statistically significant.

This result is actually informative *despite* the other data quality issues in this run, because the backtest depends only on (a) whether the game went over or under the posted total and (b) the B2B flag — neither of which is affected by the odds-parsing problem. The conclusion that a simple B2B under-betting strategy has no edge is therefore reliable on the current data.

Verdict: The backtest provides no support for a profitable B2B under-betting strategy on this sample. Net ROI = –4.94%, Sharpe = –0.81. The mispricing detected in the VRP and subgroup tests (Section 3) does not translate into a mechanically exploitable rule at the B2B + above-median-line filter level.

7. Hypothesis Scorecard & Next Steps

Hypothesis	Test	Result	Verdict
H1 — Market calibrated ($\alpha=0$, $\beta=1$)	Mincer-Zarnowitz OLS	$\alpha=+0.515$ ($p<0.001$) $\beta=-0.002$ ($p=0.972$) Brier skill = -0.25	✗ Rejected Market is not calibrated
H2 — Aggregate VRP > 0	One-sample t-test on residuals	VRP= -0.99 , $t=2.20$ $p=0.028$	✓ Supported* *magnitude suspect
H3 — B2B games go under	OLS coefficient both_b2b, one_b2b	$\beta=+2.99$ ($p=0.216$) $\beta=-0.69$ ($p=0.525$)	✗ Not supported No significant effect
H4 — Star absence depresses scoring	OLS coefficient star_absent	$\beta=+4.16$ ($p<0.001$) Wrong sign	✗ Contaminated Odds parsing artefact
H5 — Favorite-Longshot Bias	Decile binning implied vs. realized	Bimodal p distribution (0.24 and 0.76 clusters)	■ Indeterminate Odds parsing artefact

Root cause: SBR odds parsing

The calibration, FLB, VRP magnitude, and H4 sign issues all trace back to the same root cause: the SBR loader is inconsistently assigning over vs. under odds from the two-row game format. In the SBR files, the visitor row carries the spread and the home row carries the total line — but the *odds* attached to the total (the closing price at which you can bet over or under) need to come from the correct row for the correct side. When they are swapped, a game priced at 0.50 fair probability gets assigned 0.24 or 0.76 instead, producing all the downstream distortions observed.

Priority fix

Issue	Fix
Bimodal p_over_fair (0.24 / 0.76 clusters)	Audit SBRLoader.load_sbr_file: verify that over_odds and under_odds are consistently extracted from the correct row
VRP magnitude of -0.99	Re-run after fixing odds assignment. Expect VRP to shrink to single-digit basis points for symmetric lines; any remaining bias is likely due to non-symmetric lines
star_absent sign reversal	Re-run after fixing odds assignment. Expect $\beta < 0$ consistent with the hypothesis once the residual is computed against the correct line
Sample size (1,738 vs ~4,920 expected)	Update SEASONS config to include 2019-20 and 2020-21, delete cached CSVs, re-run nba_api downloads to match seasons

Pipeline code, data, and reproducibility instructions available upon request. To replicate: place SBR Excel files in data/sbr/, then run `nba_prop_pipeline.py`.